

EX-Gaze: High-frequency and Low-latency Gaze Tracking with Hybrid Event-frame Cameras for On-Device Extended Reality

Ning Chen , Yiran Shen *, Tongyu Zhang , Yanni Yang , and Hongkai Wen 

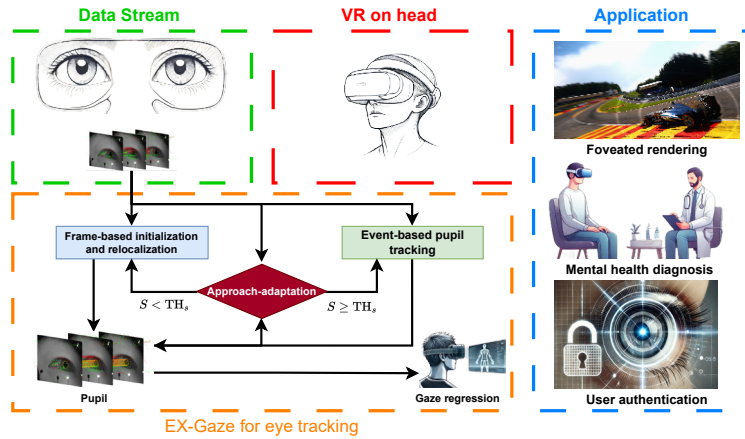


Fig. 1: EX-Gaze utilizes a combination of low-rate intensity frames and high-temporal-resolution event streams from event cameras as input. This approach enables high-frequency, real-time eye tracking for mobile extended reality applications that demand rapid and precise eye-tracking capabilities.

Abstract—The integration of gaze/eye tracking into virtual and augmented reality devices has unlocked new possibilities, offering a novel human-computer interaction (HCI) modality for on-device extended reality (XR). Emerging applications in XR, such as low-effort user authentication, mental health diagnosis, and foveated rendering, demand real-time eye tracking at high frequencies, a capability that current solutions struggle to deliver. To address this challenge, we present EX-Gaze, an event-based real-time eye tracking system designed for on-device extended reality. EX-Gaze achieves a high tracking frequency of 2KHz, providing decent accuracy and low tracking latency. The exceptional tracking frequency of EX-Gaze is achieved through the use of event cameras, cutting-edge, bio-inspired vision hardware that delivers event-stream output at high temporal resolution. We have developed a lightweight tracking framework that enables real-time pupil region localization and tracking on mobile devices. To effectively leverage the sparse nature of event-streams, we introduce the sparse event-patch representation and the corresponding sparse event patches transformer as key components to reduce computational time. Implemented on Jetson Orin Nano, a low-cost, small-sized mobile device with hybrid GPU and CPU components capable of parallel processing of multiple deep neural networks, EX-Gaze maximizes the computation power of Jetson Orin Nano through sophisticated computation scheduling and offloading between GPUs and CPUs. This enables EX-Gaze to achieve real-time tracking at 2KHz without accumulating latency. Evaluation on public datasets demonstrates that EX-Gaze outperforms other event-based eye tracking methods by striking the best balance between accuracy and efficiency on mobile devices. These results highlight EX-Gaze’s potential as a groundbreaking technology to support XR applications that require high-frequency and real-time eye tracking. The code is available at <https://github.com/Ningreka/EX-Gaze>.

Index Terms—Gaze Tracking, Event Camera, Sparse transformer

1 INTRODUCTION

Extended reality (XR), encompassing virtual reality (VR) and augmented reality (AR), is currently experiencing a surge in research and industrial investment due to its potential in immersive computing tech-

nologies [30, 31]. These technologies aim to bridge the gap between the physical and digital worlds, offering compelling and realistic digital experiences. One of the keys to achieving this is the advancement of human-computer interaction (HCI) in virtual environments. Gaze/eye tracking has emerged as a crucial HCI technique in XR, rapidly gaining adoption in cutting-edge head-mounted displays with embedded computers, including Apple Vision Pro [1], Meta Quest [3], HTC VIVE [6], Microsoft Hololens [4], ByteDance Pico [5], etc. By accurately tracking what the user is looking at, the eye tracking enhances immersion, interaction, and accessibility in XR, opening up new possibilities for innovative applications and experiences, such as foveated rendering [38], mental health diagnosis [13, 18], users’ behavior analysis [18, 36], identity authentication [26, 28], and etc.

The rapid movements of human eyes during the saccade state, with speeds of up to $700^\circ/s$ [8] in angular and accelerations of $24,000^\circ/s^2$ [9], underscore the need for gaze tracking solutions with high temporal resolution. This level of detail is crucial for applications such as mental health diagnosis [13, 18], foveated rendering [10, 38],

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and gaze authentication [28]. Traditional eye tracking methods often rely on CCD/CMOS cameras to locate the pupil region in images and then estimate the 3D gaze direction from regression model [20, 21]. However, these cameras typically operate at frame rates of only 30–50Hz, limiting the tracking frequency at such level. While the EyeLink 1000 [7] with high-speed cameras can achieve tracking frequencies of up to 1KHz, they come with significant computational, memory, and bandwidth requirements, posing significant challenges for real-time processing on embedded computers.

To reconcile the contradiction between high tracking frequency and resource consumption, event cameras have recently been introduced to track eye movement in high-frequency [11, 12, 15, 52–54]. Unlike traditional frame-based cameras, event cameras detect pixel-wise intensity changes independently, producing asynchronous and sparse event-streams in high temporal resolution instead of synchronous video frames at a low rate. Event cameras have shown superior performance on a number of applications including video generation [40], frames interpolation [46], gait recognition [47], and etc. Compared to CCD/CMOS cameras, event cameras offer several characteristics that make them promising for eye tracking on embedded computers. First, their temporal resolution can reach tens of microseconds, allowing them to capture the detailed phases of fast and brief eye movement events during saccades without motion blur. Additionally, event cameras consume significantly fewer computational, memory, bandwidth, and energy resources due to the sparse nature of their event-streams, making them ideal for implementation on resource-constrained embedded computers.

This paper focuses on achieving real-time and accurate gaze tracking at frequencies exceeding a few kilohertz using event cameras implemented on embedded computers, specifically the Jetson Orin Nano 8GB [27] to support the XR applications requiring both high temporal resolution and low tracking latency, e.g., foveated rendering. The Jetson Orin Nano, part of the Jetson family, is the smallest (70mm×45mm) and most power-efficient embedded computer, capable of running multiple neural networks in parallel. Therefore, it has the potential to be embedded in mobile XR devices in the near future. While several attempts have been made to implement eye/gaze tracking with event cameras, none have achieved the goals outlined above. For example, EBVEYE [11] and EV-Eye [54] primarily focus on collecting eye tracking datasets with event cameras and propose benchmarking methods to demonstrate the effectiveness of event-based eye tracking, rather than optimizing performance for accuracy and efficiency. Swift-Eye [52] significantly improves the accuracy of eye tracking and addresses occlusion issues caused by involuntary blinks by exploiting the high temporal resolution provided by event cameras. However, its complex design requires significant resources and is not suitable for implementation on embedded computers for real-time tracking. Similar to our work, 3ET [15] and Retina [12] aim to reduce the resource consumption of event-based eye tracking. However, they simply adopt simple neural networks to improve efficiency, which may not guarantee comparable tracking accuracy or fully utilize the computational capacity of the embedded computers' heterogeneous computing components, i.e., CPUs and GPUs.

To address the gaps mentioned above, we propose EX-Gaze, an ultrafast and high-frequency eye-tracking approach designed for embedded edge devices. By utilizing hybrid event-frame cameras such as the DAVIS346 [2], which capture both high-temporal-resolution event streams and intensity frames rich in semantic detail, EX-Gaze fully leverages the strengths of both modalities. This combination allows for high-precision tracking with low resource consumption. Additionally, EX-Gaze harnesses the computational power of the Jetson Orin Nano to run deep neural networks, delivering optimal performance in terms of tracking frequency, latency, and accuracy. The contributions of EX-Gaze are as follows:

- By leveraging the sparse nature of event-streams and the rich semantic information of video frames provided by the hybrid event-frame cameras, we have designed several bespoke components to accelerate processing and improve the accuracy of eye tracking simultaneously.
- Extensive evaluations validate the design of EX-Gaze and demonstrate its significantly improved trade-off between accuracy and efficiency compared to existing event-based eye tracking approaches.

2 RELATED WORK

In this section, we will review the relevant work to the frame-based and event-based eye tracking in the literature.

Frame-based eye tracking. The frame-based eye tracking methods rely on the recording of eye images and track the location of the eye region according to the appearance of the eyes in images. For example, DeepVOG [51] uses a U-Net-based network to segment the pupil region and fit an elliptical contour based on the segmentation results. RITnet [14] optimized the U-Net network to reduce the amount of computation and parameters and introduced a novel loss function specifically tailored for pupil segmentation. The EyeNet [49] method is the first to use a single model to simultaneously perform eye segmentation, IR LED glint detection, pupil and cornea center estimation. The above methods that fit an elliptical contour based on pupil and iris segmentation often suffer from performance degradation when the pupil or iris is occluded. This is because the accuracy and completeness of the segmentation directly impact the effectiveness of the fitting process. To address these limitations, EllSeg [29] proposed a new eye tracking framework, which introduces an additional branch to the segmentation model, enabling it to regress the ellipse axis and orientation parameters from the bottleneck representations. Additionally, the Center of Mass loss was proposed to enhance the model's ability to extract pupil and iris centers. Wang et al. [48] improves on EllSeg by using adversarial learning to train an edge extraction network to generate high-quality edges of the eyes in the image and input both the original image and the edge image into an edge-guided segmentation and fitting network (ESF-Net) to obtain the segmentation of the eyes and fit the ellipses of the pupil and iris. In addition, they further optimize the obtained ellipse parameters based on the segmentation through parameter search methods. Feng et al. [19] proposed an event-driven ROI (region of interest) prediction method to reduce computation. This method is based on the intuition that changes in the background are less significant than those caused by the movement of the foreground eye parts. By emulating an event camera, the method extracts the temporal correlation across frames and predicts the ROI.

Event-based eye tracking. Constrained by hardware limitations and imaging principles, frame-based methods are hard to achieve high-frequency tracking. To address these limitations, several event-based methods have emerged to achieve higher-frequency eye tracking. EBVEYE [11], the first event-based eye tracking method, leverages the characteristics of both event and frame data to achieve ultra-high-frequency eye tracking. It employs morphological methods to extract the pupil's edge from frames for fitting an elliptical contour and filtering events based on their distance to the elliptical contour for updating the contour. EV-Eye [54] and Swift-Eye [52] also adopt a frame-event hybrid approach. EV-Eye utilizes U-Net to segment the pupil within frames. Events are then filtered based on their distance to the pupil's center of mass, and the corresponding translations are calculated to update the pupil's position. Swift-Eye utilizes event data for interpolation to obtain high-frequency frames and then applies frame-based methods for pupil detection. Notably, it employs a rotated object detection method instead of a segmentation method to obtain the pupil ellipse. To avoid relying on additional data formats, some researchers are exploring purely event-based eye tracking methods. A significant challenge in purely event-based eye tracking lies in extracting meaningful information from scenes with minimal changes. To overcome this challenge, Stoffregen et al. [44] designed an illumination scheme, Coded Differential Lighting, that introduces artificial brightness changes into the scene. This approach generates events with a consistent pattern,

facilitating effective glint detection and tracking. Other purely event-based methods typically employ event accumulation to achieve eye tracking. For instance, both E-Track [32] and E-Gaze [33] utilize a fixed number of events as input to the method. The distinction lies in the segmentation approach: E-Track employs U-Net to segment events belonging to the pupil, while the E-Gaze method utilizes morphological segmentation. To reduce computational complexity, both E-Track and E-Gaze employ a simplified approach for updating the pupil's position once it has been successfully detected. Instead of segmenting events across the entire frame for each update, these methods restrict their analysis to a toroidal band based on the previously estimated pupil position. 3ET [15] introduced a sparse Change-Based Convolutional Long Short-Term Memory (CB-ConvLSTM) model for event-based eye tracking, achieving significant computational efficiency compared to the standard ConvLSTM-based model. The method utilizes events within a fixed-length time window as input to the network. However, this accumulation method fails to provide the network with consistent and effective eye features for prediction, resulting in suboptimal tracking accuracy for 3ET. A most recent work, Retina [12], utilizes a spiking neural network (SNN) to achieve low-latency eye tracking. While SNNs hold promise for substantial computational efficiency compared to traditional Artificial Neuron Network (ANN), their practical implementation demands specialized neuromorphic processors. On commonly used processors like CPUs and GPUs, SNNs demonstrate suboptimal performance. FAPNet [35], a purely event-based method, represents event data as point clouds and integrates the point-based architecture PEPNet [41] with LSTM. To ensure adequate information for detection, the preprocessing stage employs Frequency Adaptive Window Expanding, dynamically adjusting the window size from 10ms to 100ms to capture sufficient events. Nonetheless, the minimum window size of 10ms limits its tracking frequency to 100Hz and the number of parameters and FLOPs of FAPNet are significantly larger than ours.

3 SYSTEM DESIGN

3.1 System Overview

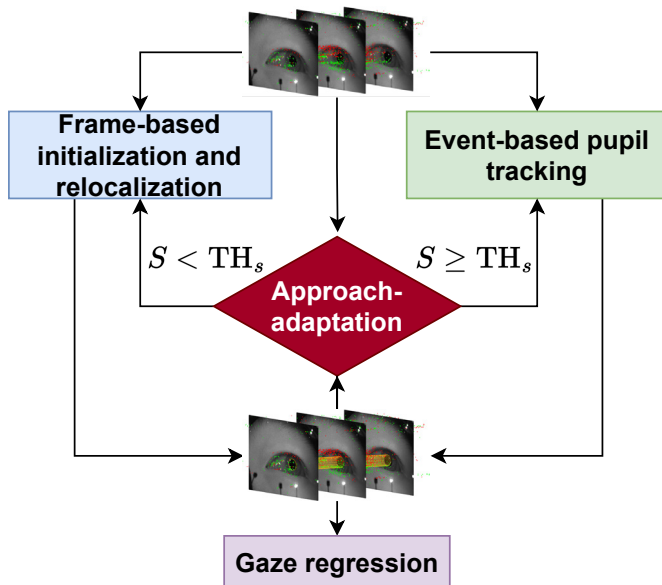


Fig. 2: System overview of EX-Gaze

As depicted in Figure 2, EX-Gaze can be divided into four major modules: the frame-based initialization and relocalization module, the event-based pupil tracking module, the approach-adaptation module, and the gaze regression module. EX-Gaze is designed to work with hybrid event-frame cameras, such as the DAVIS346 [2], which record both event-streams and video frames simultaneously. This setup allows EX-Gaze to leverage the high-temporal resolution of event-streams to

improve tracking frequency, while also exploiting the rich semantic information contained in video frames to enhance tracking accuracy.

EX-Gaze begins with the frame-based initialization and relocalization module, which is responsible for locating the initial region of the eyes and pupils at the start and relocalizing the pupil region if the event-based tracking is lost. After determining the initial locations of the eyes and pupils, the event-based pupil tracking module is activated to track the movement of pupils at a high frequency, such as 2KHz in this paper. This module leverages the sparse nature of the event-stream and the lightweight design of deep neural networks to achieve low computation time on GPUs of the Jetson Orin Nano 8GB, typically less than 0.4ms.

Next, the approach-adaptation module calculates the tracking error based on the gradient similarity between the pupil boundaries from the tracking result and the eye frame. If the discrepancy of the event-based tracking exceeds a predefined threshold, this module triggers the frame-based relocalization.

Finally, the parameters of the pupil region are fed into a gaze regression module, which utilizes a common polynomial regression model to map the location of the pupil region in the image domain to the point of gaze (PoG) on the screen.

3.2 Eye Model

We first introduce the parametric model of eyes and pupils to be tracked by EX-Gaze, whose parameters are updated whenever a slice of event-stream or a video frame is processed to obtain the new location of the eyes or pupil regions.

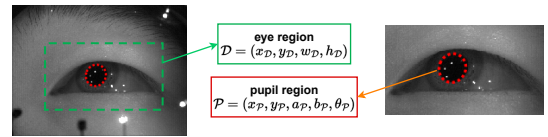


Fig. 3: An example of near-eye image labeled with parameters of eye region (left) and the cropped eye region labeled with the parameters of pupil region (right).

As shown in Figure 3, EX-Gaze focuses on the near-eye eye tracking setting which is also adopted by most of the VR/AR headsets. Our eye model considers two important regions associated with eye tracking: the eye region and the pupil region. The eye region is defined as a bounding box containing the whole eyeball, canthi, eyelashes, and other relevant features. It can be described as a rectangular area $\mathcal{D}$ that encompasses all visible parts of the eye and can be parameterized as,

$$\mathcal{D} = (x_{\mathcal{D}}, y_{\mathcal{D}}, w_{\mathcal{D}}, h_{\mathcal{D}}) \quad (1)$$

where $(x_{\mathcal{D}}, y_{\mathcal{D}})$ denote the location of the eye region and $(w_{\mathcal{D}}, h_{\mathcal{D}})$ are the width and height of the bounding box. Near-eye tracking relies on the fact that the location of the eye region in the image domain remains relatively fixed once a specific user puts on the VR/AR device. Therefore, the rectangle area of the eye region can be cropped once at the initialization stage. This cropped area serves as a small-size input for the subsequent processing steps, accelerating the frame-based pupil relocalization.

Then the right part in Figure 3 shows an example of closed-up eye with annotated pupil region $\mathcal{P}$, parameterized as an ellipse, i.e.,

$$\mathcal{P} = (x_{\mathcal{P}}, y_{\mathcal{P}}, a_{\mathcal{P}}, b_{\mathcal{P}}, \theta_{\mathcal{P}}) \quad (2)$$

where $(x_{\mathcal{P}}, y_{\mathcal{P}})$ are the coordinates of the pupil center, $a_{\mathcal{P}}$ and $b_{\mathcal{P}}$ are the lengths of the long and short axes of the ellipse representing the pupil, and $\theta_{\mathcal{P}}$ is the orientation of the ellipse. The primary objective of EX-Gaze is to determine these parameters of the pupil region using information from both the video frames and event-streams efficiently.

3.3 Frame-based Initialization and Relocalization

As previously introduced, EX-Gaze utilizes the DAVIS346, a hybrid event-frame platform, to capture eye movements in close proximity.

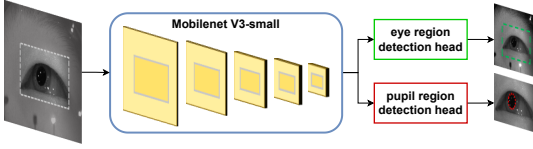


Fig. 4: Frame-based initialization and relocalization

The low-rate near-eye video frames collected along with the event-streams contain abundant information around human eyes, making them valuable for extracting accurate eye and pupil regions. However, due to the low frame rate (25fps for DAVIS346) and the higher resource consumption of processing video frames compared to event-streams, the frame-based eye and pupil region detection is triggered only occasionally: at the once-off initialization stage and when relocalization is needed due to the loss of event-based tracking.

As depicted in Figure 4, the backbone of the frame-based initialization and relocalization module is based on MobileNetV3-small [23], a lightweight neural network designed for mobile and embedded vision applications. It primarily consists of MBConvBlocks, which are efficient convolution operations. MobileNetV3 is used to extract features from near-eye images, and its output features are fed into two detection heads, similar to that used in RetinaNet [34], for detecting the eye region and pupil region, respectively.

Given that the eye region remains relatively fixed under the near-eye setting of the head-mounted device, the eye region detection head only needs to run during the initialization stage when the user puts on the device. Subsequently, the input of the MobileNetV3 is reduced from the whole near-eye image to the significantly smaller eye region cropped from the original image, accelerating the frame-based pupil detection. In the relocalization stage, the frame-based module is activated again to estimate the parameters of the pupil region using the cropped eye region as input. This occurs when the high-speed event-based eye tracking fails to produce acceptable results.

3.4 Event-based Pupil Tracking

Constrained by the limited frame rate, a frame-based approach alone cannot achieve high-frequency eye tracking with a refresh rate exceeding thousands of hertz to support applications requiring detailed eye movement analysis in high-temporal resolution, such as foveated rendering and mental health diagnosis. Thanks to the hybrid design of the DAVIS346, which allows for the simultaneous capture of both video frames with abundant semantic information and high-temporal resolution event-streams, the event-streams can be exploited to complement the frame-based approach. This integration enables the achievement of high-accuracy and high-frequency eye tracking.

However, achieving high-frequency eye tracking in real-time poses a significant demand on computation speed, which can be challenging for resource-constrained, small-size embedded computers. For example, to support a tracking frequency exceeding 2KHz, the entire processing time to obtain each tracked result, from data acquisition to gaze output, should be limited to less than 0.5ms to avoid accumulated tracking latency. Moreover, as the DAVIS346 can only generate 25 frames per second, realizing high-frequency tracking in real-time depends heavily on the processing speed of the event-based pupil tracking rather than the frame-based approach. Thus, the core of EX-Gaze is an efficient event-based pupil tracking module designed for embedded computers (e.g., Jetson Orin Nano). This module enables ultrafast eye tracking at extremely high frequencies (e.g., 2KHz) without accumulated latency.

Recognizing that reducing the input size is an effective measure to accelerate processing speed, the sparse nature of event-streams can be leveraged to improve the efficiency of eye tracking. One existing solution to exploit such sparsity is to adopt sparse convolutional neural networks (sparse-CNNs) [22]. However, sparse-CNNs are yet to be optimized properly and can be difficult to implement efficiently with Compute Unified Device Architecture (CUDA) for processing in parallel on GPUs, which is more suitable for accelerating the processing of traditional CNNs on dense input. This is because the sparse input

of event-streams can have non-structural dynamic data layouts. As a result, sparse-CNNs often require additional overhead to manage the sparse input, which can cancel out the potential gains from sparsity.

Thus, to leverage the sparse nature of event-streams effectively, we propose a Sparse Event Patches Transformer (SEPT) as the backbone of the event-based pupil tracking module. SEPT, as illustrated in Figure 5, starts by taking a few patches of sparse events around the edge of the pupil region, triggered by the movement of the eyeballs, as input. These sparse patches are then processed through a series of shared convolutional layers to extract local features from each event patch, which form patch embeddings. To preserve the positional relation between patches, learnable positional embeddings are adopted. Finally, a transformer encoder with pooling and fully connected layers is applied to fuse the local features extracted from each patch. This process calculates the offset caused by the instantaneous eye movement from the previous tracking result to the current timestamp, which is used to decode the current parameters of the pupil region.

3.4.1 Sparse Event-patch Representation

To be compatible with the SEPT module, we propose a sparse event-patch representation for the sparse and non-structural event-streams, as shown in Figure 6. We denote $\mathcal{E}_{t_0 \rightarrow t_n}$ as a very short slice of event-stream, e.g., 5ms, recording the eye movement between the timestamps t_0 and t_n . As shown on the left of the figure, the event-stream consists of a number of positive (green dots) and negative (red dots) events triggered by the eye movement, and a short slice with N events can be represented as,

$$\mathcal{E} = \{e_i\}_{i=1:N} = \{x_i, y_i, t_i, p_i\}_{i=1:N} \quad (3)$$

where (x_i, y_i) are the pixel coordinates, t_i is the timestamp and p_i is the polarity of the i_{th} event.

Then the events in the slice of event-stream are accumulated according to their coordinates and polarities, i.e.,

$$I(x, y, p) = \sum_{i \in N} \delta(x - x_i, y - y_i, p - p_i) \quad (4)$$

where δ represents the Kronecker function, which outputs ‘1’ for a specific combination of coordinates (x_i, y_i) and polarity p_i , and ‘0’ otherwise. Intuitively, it effectively counts the number of positive and negative events on each pixel respectively. By applying the Kronecker function δ , we can convert the slice of event-stream into an event-image with two channels, distinguished by the polarities of the events.

According to the imaging principle of event cameras, events triggered by eye movements are mainly distributed around parts with distinctive edges, such as the edge of pupils, iris, and eyelashes. Therefore, in event-based pupil tracking, we only consider a few events located in proximity to the edge of the pupil region. This sparsity constraint benefits the eye tracking system in both efficiency and accuracy by excluding irrelevant events, such as noises and irrelevant movements, from processing.

To exploit the benefit derived from the sparsity of events corresponding to pupil movement, we design a sparse event-patch representation to significantly reduce the input size for the following deep neural networks, which are well-optimized for efficiency on CUDA. As shown in the middle of Figure 6, a few rectangular event-patches, typically 8 in this paper, are extracted around the edge of the pupil region. These patches serve as the input to traditional convolutional layers, which are well-optimized for efficient processing on the GPUs of embedded computers. The locations of the event-patches are determined by the ellipse obtained from the latest tracking result. For example, for the current timestamp t_1 , the ellipse from the latest tracking result can be parameterized as,

$$\mathcal{P}^{t_0} = (x_{\mathcal{P}}^{t_0}, y_{\mathcal{P}}^{t_0}, a_{\mathcal{P}}^{t_0}, b_{\mathcal{P}}^{t_0}, \theta_{\mathcal{P}}^{t_0}) \quad (5)$$

As the targeted tracking frequency is 2KHz, the displacement of the pupil region from the last tracking timestamp t_0 to the current timestamp t_1 (i.e., 0.5ms) is not significant. Therefore, as shown in

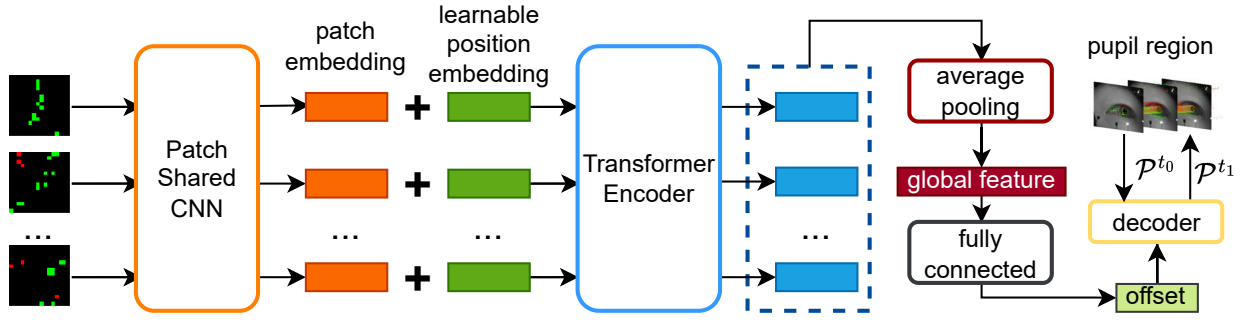


Fig. 5: Sparse event-patch transformer for event-based pupil tracking

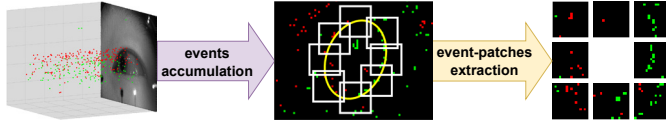


Fig. 6: Sparse event-patch representation

the middle of Figure 6, we place square patches along the edge of the ellipse (in yellow) obtained from the last tracking result to generate the sparse event-patch representation. Specifically, M points (8 points in this paper) are sampled on the ellipse in a counter-clockwise direction with a radian interval of $\frac{2\pi}{M}$ as the centers of the patches, and their side lengths are equal (16 pixels in this paper). Based on the range of these square patches, we extract the event-patches from the accumulated event-stream $I(x, y, p)$ to form the input of the following deep neural networks for eye tracking.

Tracking in Adaptive Frequency. As mentioned before, the movement of eyeballs can be categorized into three major states: saccade, smooth pursuit, and fixation, each with significantly different moving speeds. Saccadic movements require tracking updates at a significantly high frequency, while the eyeballs rarely move during fixation states. Therefore, high-frequency updates during fixation states are neither necessary nor accurate, considering the noisy nature of event cameras caused by unstable hardware and changing lighting conditions. The noisy events irrelevant to the movement of the eye may dominate in the patches during fixation states, leading to significant tracking errors in EX-Gaze when the eyeballs do not move. Conversely, a considerable number of events will be generated within a brief period if the eyeballs move quickly. Therefore, an adaptive strategy is required according to the speed of event generation to accommodate the significantly distinctive movement states in eye tracking.

Specifically, we consider two important factors to determine if a tracking update is needed during the accumulation of event-patches: the total number of accumulated positive and negative events in square patches (e^c) and the period of accumulation (t^c). For the current timestamp t , an event-based tracking update will be triggered when

$$e_{(t-t_c) \rightarrow t}^c > TH_{ec} \quad \& \quad t^c < TH_{tc} \quad (6)$$

where $e_{(t-t_c) \rightarrow t}^c$ is the total number of events accumulated from $t - t_c$ until now (t), and TH_{ec} and TH_{tc} are the thresholds for the accumulated events and the accumulation period to trigger the event-based tracking. In our system implementation, TH_{ec} is set to 50 and TH_{tc} is set to 5ms. To achieve the target tracking frequency of 2KHz, for each 0.5ms, the triggering conditions above will be checked.

3.4.2 Sparse Event-Patch Transformer (SEPT)

As depicted in Figure 5, after the events are accumulated and converted to event-patches, the Sparse Event-Patch Transformer (SEPT) is used to compute the offset of the pupil region from the last tracking result to the current timestamp to support real-time pupil tracking in high-frequency.

Inspired by vision-transformer [16], the backbone of SEPT comprises two major components: a patch-shared convolutional neural network (CNN) for local feature extraction and a transformer-encoder for fusing the local features from different event-patches. Specifically, the patch-shared CNN consists of one convolutional layer followed by three consecutive Fused-MBConvBlocks [45]. To save resources and promote parallel computing on embedded GPUs, the coefficients of the convolutional layers are shared across different event-patches. As the sparse event-batches are distributed around the pupil region, their geometric relations are also important for tracking the whole pupil region accurately. Thus, after the local features of different patches are extracted from the Fused-MBConvBlocks, learnable positional embedding [16] is adopted to preserve the relative positions between the patches. Then the outputted feature embeddings are fed to a transformer encoder with lightweight separable self-attention mechanism [37] for feature fusion. The global feature vector is obtained from the average pooling operation on the fused features and serves as the input of a fully connected layer to calculate the offset of the pupil from the last tracking result at timestamp t_0 to the current timestamp t_1 . The definition of the offset in our algorithm is similar to but slightly different from that in rotated object detection (mmdrotate) [50, 55] in which the offset is the difference between the anchor box and bounding box of the detected object. Specifically, the outputted offset from SEPT can be parameterized as,

$$\mathcal{O}_{\mathcal{P}}^{t_0 \rightarrow t_1} = (x_{\mathcal{P}}^{t_0 \rightarrow t_1}, y_{\mathcal{P}}^{t_0 \rightarrow t_1}, a_{\mathcal{P}}^{t_0 \rightarrow t_1}, b_{\mathcal{P}}^{t_0 \rightarrow t_1}, \theta_{\mathcal{P}}^{t_0 \rightarrow t_1}) \quad (7)$$

where $(x_{\mathcal{P}}^{t_0 \rightarrow t_1}, y_{\mathcal{P}}^{t_0 \rightarrow t_1})$ are the offsets of the center point of the pupil region, $a_{\mathcal{P}}^{t_0 \rightarrow t_1}$ and $b_{\mathcal{P}}^{t_0 \rightarrow t_1}$ are the offsets of length of the long and short axes, and $\theta_{\mathcal{P}}^{t_0 \rightarrow t_1}$ is the offset of the orientation, which indicate the changes of the pupil between two consecutive tracking timestamps.

Finally, the last tracking result $\mathcal{P}^{t_0}$ obtained at timestamp t_0 and the offset estimated from SEPT are fed to a decoder to determine the parameters of the pupil region at current timestamp t_1 , which is defined as,

$$\begin{aligned} x_{\mathcal{P}}^{t_1} &= x_{\mathcal{P}}^{t_0} - x_{\mathcal{P}}^{t_0 \rightarrow t_1} \times w_{\text{anchor}}, \\ y_{\mathcal{P}}^{t_1} &= y_{\mathcal{P}}^{t_0} - y_{\mathcal{P}}^{t_0 \rightarrow t_1} \times h_{\text{anchor}}, \\ a_{\mathcal{P}}^{t_1} &= \frac{a_{\mathcal{P}}^{t_0}}{e^{a_{\mathcal{P}}^{t_0 \rightarrow t_1}}}, \\ b_{\mathcal{P}}^{t_1} &= \frac{b_{\mathcal{P}}^{t_0}}{e^{b_{\mathcal{P}}^{t_0 \rightarrow t_1}}}, \\ \theta_{\mathcal{P}}^{t_1} &= \theta_{\mathcal{P}}^{t_0} + \theta_{\mathcal{P}}^{t_0 \rightarrow t_1} \end{aligned} \quad (8)$$

where w_{anchor} and h_{anchor} are the width and height of the anchor box defined in mmdrotate [50, 55]. With the decoding operation, the final event-based tracking result can be obtained. It is worth noting that, the highest temporal resolution of the eye tracking is determined by the interval from t_0 to t_1 , which is 0.5ms in our real-world implementation of EX-Gaze.

3.5 Event-frame Approach-adaptation

The event-based and frame-based approaches each have their own advantages and issues due to their unique characteristics. The event-based approach provides high-temporal resolution and requires low computation on embedded computers. However, it may not guarantee satisfactory accuracy for long-term tracking due to its lack of semantic information and noisy nature. On the contrary, the video frames are semantically rich, and pupil detection on video frames is accurate and robust. However, its dense input requires more computation than the event-based approach, thus, it cannot support real-time eye tracking at high frequency.

To fully leverage the advantages of both event-based and frame-based approaches while mitigating their limitations, we propose an approach-adaptation scheme to guide the EX-Gaze in switching between these methods effectively. For high-frequency eye tracking, the EX-Gaze primarily employs event-based pupil tracking but switches to frame-based tracking occasionally when the tracking result deviates significantly from the correct location. One common approach to assess this deviation is to compare the frame-based detection with the event-based tracking results for each newly arrived frame. However, this method introduces unnecessary computation and may lead to accumulated delays in high-frequency tracking.

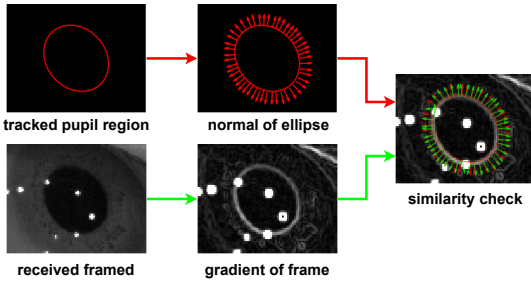


Fig. 7: Gradient-based template matching

To bridge the gap above, we propose a lightweight approach-adaptation scheme based on the gradient-based edge matching [43] to check the tracking accuracy on each incoming frame efficiently without performing the computationally intensive frame-based pupil detection. As shown in Figure 7, when a new video frame F_t is recorded at timestamp t , the current event-based pupil tracking result, parameterized as an ellipse,

$$\mathcal{P}^t = (x_{\mathcal{P}}^t, y_{\mathcal{P}}^t, a_{\mathcal{P}}^t, b_{\mathcal{P}}^t, \theta_{\mathcal{P}}^t) \quad (9)$$

It is evident that when pupil tracking is accurate, the ellipse should closely match the high-contrast edge between the pupil and iris region, and the high-contrast edge will result in relatively large gradient values. Based on these observations, we first sample K discrete points uniformly along the edge of the ellipse and calculate their corresponding set of normal vectors ($\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3, \dots, \mathbf{n}_K$) according to the shape of the ellipse. Next, we estimate the image gradients around these points $\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3, \dots, \mathbf{g}_n$ using the Sobel operator [17, 42] based on the values of neighboring pixels. Finally, we compute the similarity between the ellipse normals and the image gradients at the sampled points as follows,

$$S = \frac{1}{K} \sum_{i=1}^K \frac{\langle \mathbf{n}_i, \mathbf{g}_i \rangle}{\|\mathbf{n}_i\| \cdot \|\mathbf{g}_i\|} \quad (10)$$

According to the definition above, the values of similarity S range from 0 to 1, where a value closer to 1 indicates higher similarity between the shape of the ellipse and the distribution of image gradients at the sampled points. During event-based pupil tracking, the similarity is regularly checked when a new frame is acquired, and frame-based relocalization will be triggered when the similarity falls below a predefined threshold TH_s ($\text{TH}_s = 0.8$ in the following evaluation section).

Based on our evaluation on Jetson Orin Nano 8GB, the computation of sparse normals and image gradients is significantly more efficient than frame-based pupil detection, achieving nearly a 60-fold

improvement in computation time (0.02ms vs. 1.17ms). Therefore, EX-Gaze can adaptively switch to the appropriate approach based on real-time prediction of the tracking accuracy with only minimal overhead.

It is worth noting that, as shown in the bottom left of Figure 7, significant interference around the pupil region in the images, such as glints, may occur due to reflections from extra lighting used to enhance the contrast of the pupil region and improve image quality. To address this interference, the similarity computation excludes points with outlier gradients, which are determined by checking the distribution of gradients and excluding values that are far from the mean of the distribution, for example, those more than 2 times the standard deviation away from the mean.

3.6 Gaze Calibration and Regression

So far, we have described how to detect and track pupil regions with frames and slices of event-streams in the camera domain. To determine what the user is looking at on the screen, a polynomial regression model is adopted to build the correspondence between the location of the pupil in the camera domain and the point of gazes (PoGs) on the screen, the n -order polynomial regression model can be expressed as,

$$\begin{aligned} x_{\text{PoG}} &= a_0 + \sum_{u=1}^n \sum_{v=0}^u a_{(v,u)} x_{\mathcal{P}}^{u-v} y_{\mathcal{P}}^v \\ y_{\text{PoG}} &= b_0 + \sum_{u=1}^n \sum_{v=0}^u b_{(v,u)} x_{\mathcal{P}}^{u-v} y_{\mathcal{P}}^v. \end{aligned} \quad (11)$$

These equations represent the PoG in the horizontal and vertical directions (x_{PoG} and y_{PoG}) as functions of the pupil center coordinates ($x_{\mathcal{P}}$ and $y_{\mathcal{P}}$), where $a_{(v,u)}$ and $b_{(v,u)}$ are the regression coefficients for different orders of the center coordinates, determined in the user-dependent calibration stage.

Regression model calibration. As the kappa angles and cornea radius of eyes vary among users, the regression model should be calibrated for each user during the initialization stage before use. To calibrate the model, the screen is divided evenly into multiple regions. A stimulus point appears at the center of each region sequentially, and the user is asked to stare at the stimulus point for a while to record the correspondence between the coordinates of the pupil center in the camera domain and the PoGs on the screen. With these correspondences, the regression coefficients can be estimated easily. After the calibration stage, PoGs can be computed with the pupil tracking results according to Eq. (11). In this paper, we focus on the estimation of Points of Gaze (PoGs) as a representative example to demonstrate the complete gaze tracking process. Our approach is inherently flexible and can be readily adapted to 3D gaze estimation by employing an appropriate regression model to map the pupil center to the 3D gaze direction. Importantly, our primary contribution—pupil tracking within the camera domain—is independent of the calibration methodologies specific to various devices. This makes our approach seamlessly extendable to VR, AR, or MR devices, provided the respective calibration processes are applied.

4 EVALUATION

In this section, we conduct extensive experiments to evaluate the accuracy and efficiency of EX-Gaze for eye tracking using two public eye tracking datasets and a real-world implementation on Jetson Orin Nano 8GB. We begin by describing the public datasets used for quantitative evaluation. Next, we present the implementation details and evaluation metrics. Finally, we conduct quantitative evaluations and analyze the results to validate the design of EX-Gaze, demonstrating its superior performance compared to state-of-the-art approaches.

4.1 Dataset preparation

Dataset description. To evaluate the accuracy of EX-Gaze, we utilized two publicly available eye tracking datasets. The primary dataset is the EV-Eye public dataset [54], which we enhanced with additional labeling. This dataset captures eye movements using two DAVIS346 event cameras while the user's head is fixed with a headrest, simulating a

near-eye gaze tracking scenario. It includes grayscale image sequences captured at a lower frame rate (25fps) and event streams generated at a high temporal resolution, collected from 48 participants across four different sessions, covering various gaze movement states (fixation, saccade, and smooth pursuit). By leveraging the hybrid event-frame modalities, we fully exploit the advantages of semantic information in video frames and the high temporal resolution of event streams to achieve high-precision, high-frequency eye tracking.

Since event cameras are not yet available on real VR devices, we employed v2e [24], a widely adopted event-stream simulation approach, to generate an event-based eye tracking dataset from OpenEDS2020 [39], which was collected with VR headsets. This allowed us to demonstrate the efficacy of EX-Gaze in a VR environment. OpenEDS comprises high-resolution near-eye video frames captured at 100 fps in VR settings. From this dataset, we selected 1-second segments from each of the 200 sequences, focusing on those containing eye movements, to create a subset of 200 1-second eye movement sequences for the evaluation of our eye tracking system.

Dataset annotation and preprocessing. The EV-Eye dataset provides annotations for the pupil regions of 9,011 near-eye images. However, since these images were selected discretely, they are not suitable for evaluating continuous eye tracking accuracy. To address this limitation, we supplemented the existing annotations by labeling the pupil regions in additional sequences of near-eye images from each session of every subject. This resulted in a total of 19,270 near-eye images grouped into 384 continuous sequences, which were annotated for tracking evaluation.

For the OpenEDS2020 dataset, which provides pupil mask annotations for evaluating eye tracking methods, only video frames are available. To generate an event-based dataset from OpenEDS2020 for evaluating event-based eye tracking approaches, we first increased the frame rate of each 1-second segment by 500 times using Super-SloMo [25] for frame interpolation, providing sufficient temporal resolution for event-stream simulation. Then, we applied the video-to-events generator, v2e [24], to the high-frame-rate video sequences to synthesize corresponding event streams between each consecutive frame.

Training and testing split. For the EV-Eye dataset, we designed a user-independent split to demonstrate the generalization capability of EX-Gaze across different users. Specifically, video frames from the same user were not used in both the training and testing phases. Frames and continuous sequences from users 1 to 36 were allocated for training, while frames from users 36 to 48 were reserved for testing. To ensure a fair comparison, all competing methods used in the evaluation followed this same protocol. For evaluation on the OpenEDS dataset, models trained on the EV-Eye dataset were directly applied without fine-tuning, showcasing the system's generalization across datasets.

Training augmentation. To enhance the robustness of EX-Gaze against noise and inaccuracies during event-based continuous tracking, we introduce random perturbations on the estimated pupil parameters. This includes random translation, rotation, and scaling of the ellipse during the training stage. This approach enables the event-based tracking to accommodate inaccuracies and interference to some extent.

4.2 Evaluation Metrics

Following EV-Eye [54], we adopt three evaluation metrics to assess the accuracy of pupil tracking: the intersection over Union (IoU), F1 score, and pixel error,

- IoU measures the overlap between the predicted and annotated pupil regions $\mathcal{P}$ and $\mathcal{G}$:

$$IoU(\mathcal{P}, \mathcal{G}) = (\mathcal{P} \cap \mathcal{G}) / (\mathcal{P} \cup \mathcal{G}) \quad (12)$$

The value of IoU ranges from 0 to 1, with higher values indicating better tracking accuracy.

- F1 score is another commonly used metric for evaluating tracking accuracy. It is computed as:

$$F1(\mathcal{P}, \mathcal{G}) = (2|\mathcal{P} \cap \mathcal{G}|) / (|\mathcal{P}| + |\mathcal{G}|) \quad (13)$$

Its value is also from 0 to 1, with higher values indicating higher similarity between the predicted and annotated pupil regions.

- Pixel error (P_e) directly measures the Euclidean distance in pixels between the centers of the predicted and annotated pupil regions. Lower pixel error values indicate more accurate tracking. P_e is calculated as,

$$P_e = \sqrt{(x_{\mathcal{P}} - x_{\mathcal{G}})^2 + (y_{\mathcal{P}} - y_{\mathcal{G}})^2} \quad (14)$$

To assess if high-frequency eye tracking can be achieved in real-time on-device (i.e., Jetson Orin Nano), we also evaluate the computation time of different competing tracking methods with real-world implementation.

4.3 Ablation study

To validate the design and parameter settings of EX-Gaze, we first conduct a series of ablation studies on a number of key parameters using EV-Eye dataset, including the threshold of the number of accumulated events (TH_{ec}) to trigger the event-based tracking and the threshold of the similarity (TH_s) that controls the adaptation between frame-based and event-based approaches. In the ablation studies, the model training and estimation adhere to the described training and testing split, consistent with the subsequent evaluations. Accuracy is measured by comparing the estimated pupil region with the manually annotated ground-truth labels and the parameters are selected based on the accuracies on the test set of EV-Eye dataset.

4.3.1 Threshold of Accumulated Events

As discussed in Section 3.4.1, accumulating a sufficient number of events less than a short period of accumulation TH_{ec} ($TH_{ec} = 5ms$ for EX-Gaze) before running the event-based tracking backbone is crucial for enhancing inference accuracy and suppressing noise-induced artifacts. This strategy allows for adaptive updates of the pupil location, as the accumulation rate is proportional to the speed of eye movement. By doing so, unnecessary computations can be eliminated, leading to better utilization of the computation, energy, and bandwidth resources of embedded computers. The choice of the threshold for the accumulated events, TH_{ec} (i.e., the number of events accumulated to trigger a tracking event), impacts both tracking accuracy and tracking frequency and therefore needs to be carefully tuned.

TH_{ec}	IOU $\uparrow$	F1 Score $\uparrow$	Pixel error $\downarrow$
30	0.877	0.931	1.240
40	0.878	0.932	1.210
50	0.880	0.933	1.183
60	0.880	0.933	1.194

Table 1: The accuracy of tracking for different thresholds of accumulated events (TH_{ec})

To determine the optimal value for TH_{ec} , we evaluated the accuracy of EX-Gaze in terms of IoU, F1 score, and pixel error using different values of TH_{ec} , ranging from 30 to 60 events.

The results in Table 1 show that EX-Gaze achieves the highest accuracy for all evaluation metrics when TH_{ec} is set to 50, indicating that accumulating 50 events around the eye region within a short period (i.e., 5ms) as the condition for triggering the event-based eye tracking provides the best performance.

Therefore, according to the settings above, during event-based tracking, for each 0.5ms (corresponding to 2KHz tracking frequency), EX-Gaze will check the timestamps of the last 50 accumulated events around the eye region if their span of time is less than 5ms. If yes, the following event-based tracking will be triggered; otherwise, it will be skipped until the next 0.5ms. If the tracking at the current timestamp is skipped, the instant tracking frequency (inverse of the time interval between two tracking updates) will be less than 2KHz. To demonstrate the highest tracking frequency is achieved during the fast-moving stage, we count the number of events for each 5ms time window, and the

corresponding distribution of the number of events of EV-Eye dataset is presented in Figure 8.

From the distribution of the number of accumulated events, we can observe that over 18% of the time, EX-Gaze runs at its highest frequency (the number of accumulated events is over 50). Considering that high temporal resolution is not needed for slow eye movement, which will generate few events, the proposed adaptive updating strategy can both guarantee high temporal resolution for tracking fast eye movement and avoid unnecessary repetitive computation for slow movement, especially when the eyes are in a fixation state.

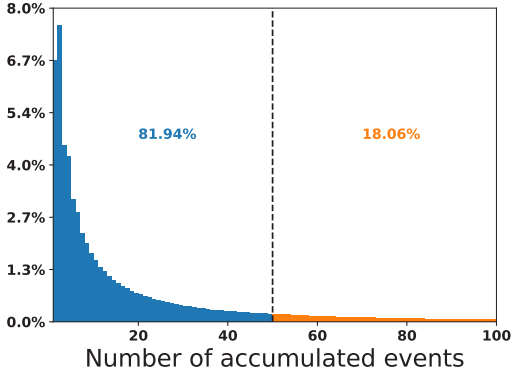


Fig. 8: The histogram of the number of accumulated events for each 5ms period drawn from EV-Eye dataset

4.3.2 Threshold of Similarity

To seamlessly switch between event-based and frame-based pupil tracking approaches as needed, we calculate the similarity between the pupil region obtained from event-based tracking and its corresponding region in the image frame. This comparison is based on an efficient gradient-based template matching method, which helps avoid the need to frequently run the computationally intensive frame-based pupil detection. Specifically, when the similarity exceeds a predefined threshold TH_s , EX-Gaze continues running the high-frequency event-based tracking in real-time. Otherwise, the frame-based pupil relocation is triggered occasionally to get the eye tracking back on track.

To determine the appropriate threshold value for the similarity metric, we evaluate the tracking accuracy and calculate the probability of triggering relocation by varying the threshold from 0.75 to 0.95 on the test set of EV-Eye dataset. The probability of relocation indicates how frequently frame-based relocation is triggered. Lowering this probability is desirable for achieving low latency. However, a lower probability of relocation can also impact tracking accuracy, as event-based tracking is inherently more challenging than the frame-based approach due to the noisy and sparse nature of event-streams.

TH_s	IOU $\uparrow$	F1 Score $\uparrow$	Pixel error $\downarrow$	probability of relocation $\downarrow$
0.75	0.855	0.920	1.454	6.3%
0.80	0.858	0.921	1.412	8.1%
0.85	0.863	0.924	1.355	10.9%
0.90	0.870	0.929	1.293	16.2%
0.95	0.875	0.931	1.259	23.6%

Table 2: The impact of similarity threshold TH_s on the accuracy and efficiency of tracking evaluated on EV-Eye dataset

The evaluation results are presented in Table 2. When comparing the results for different threshold values, we observe that increasing the threshold consistently improves tracking accuracy (measured by IoU, F1 score, and pixel error) but also leads to more frequent triggering of frame-based relocation. Therefore, the threshold should be selected based on a trade-off between accuracy and efficiency. In this paper, we

choose a threshold of 0.8 as a moderate compromise between the two factors with which, nearly 92% of the frames can be omitted to reduce the computation.

4.3.3 Evaluation on Frame-based Localization

Our event-based tracking relies on accurate frame-based pupil localization. Intuitively, inaccurate localization can negatively impact tracking performance, leading to increased relocation attempts and computational overhead. Thus, we investigate the distributions of the accuracy of the frame-based localization and its impact on the following tracking in details. We set Th_{ec} to 50 and calculated the frame-based localization accuracy (IOU_{loc}) and the event-based tracking accuracies (IOU_{track}) until the next annotated frame on the test set.

As shown in Table 3, achieving high frame-based localization accuracy significantly enhances event-based tracking accuracy, and our event-based tracking method maintains a level of accuracy comparable to the frame-based method. Notably, the first column of the table indicates that a majority of the frame-based localization results achieve relatively high IoUs, with over 97% exceeding 0.8 IOU_{loc} .

In EX-Gaze, relocation is triggered when the similarity score drops below 0.7. To explore this further, we investigated the relationship between IoUs and similarity during tracking. Our analysis revealed that when the IoU falls below 0.7, the average similarity typically drops below the predefined threshold of 0.8. This correlation suggests that maintaining an IoU of at least 0.7 (over 99% frame-based localization satisfies this conditions as shown in Table 3) in the frame-based localization step is crucial for ensuring a smooth transition to event-based tracking and minimizing the frequency of frame-based relocalizations. This becomes particularly important when employing computationally efficient models.

Range of IOU_{loc}	mean IOU_{loc}	mean IOU_{track}	Proportion
[0.9,1]	0.927	0.908	51.0 %
[0.8,0.9)	0.869	0.871	45.6 %
[0.7,0.8)	0.769	0.805	2.3 %
[0,0.7)	0.505	0.537	0.5 %

Table 3: The impact of frame-based localization accuracy on the accuracy of event-based tracking evaluated on EV-Eye dataset

4.4 Comparison with state-of-the-art approaches

We compare EX-Gaze with four state-of-the-art event-based eye tracking methods: EV-EYE [54], Swift-Eye [52], 3ET [15], and Retina [12], focusing on tracking accuracy and computation time with the two public datasets. To ensure a fair comparison, all competing approaches were trained using the same training set as EX-Gaze from EV-Eye and aimed for a 2kHz tracking frequency. For real-time performance, each method needed to complete 2,000 tracking operations in less than 1 second. Given that the regression model relies solely on pupil region centers for gaze inference, we selected pixel error as the key metric for tracking accuracy. Implementation of all methods was carried out on Jetson Orin Nano 8GB to measure the average computation time required to process each 1-second data stream.

4.4.1 Evaluation results on EV-Eye dataset

Figure 9 illustrates the trade-off between tracking accuracy and average computation time for processing each 1-second data stream for different types of eye movement, i.e., saccade and smooth pursuit from EV-Eye dataset. A preferable approach would be situated in the bottom-left quadrant, indicating low pixel error and short computation time. Our approach, as shown in the figures, ranks first in efficiency, with an average processing time of 0.39 seconds for each 1-second data stream collected under saccadic movement and 0.43 seconds under smooth pursuit. In comparison, the next most efficient method, 3ET, requires an average of 1.03 seconds for saccade and 1.01 seconds for smooth pursuit, potentially leading to slight cumulative delays, and has a much larger pixel error than our approach (24.58px v.s. 1.33px and 20.22px v.s. 1.49px for different eye movement states respectively). While

our approach achieves the second-lowest pixel error, the most accurate method, Swift-Eye, incurs a significant computational cost, taking over 1,000 seconds to process a 1-second data stream on Jetson Orin Nano. Therefore, EX-Gaze emerges as the preferred choice for high-frequency, real-time eye tracking on resource-constrained embedded computers.

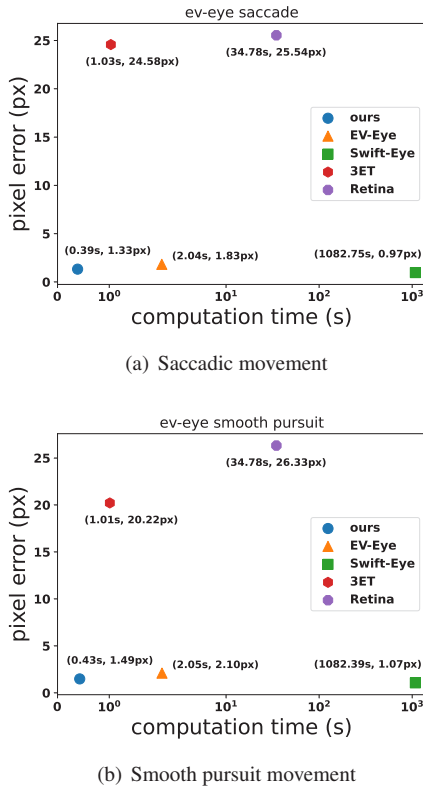


Fig. 9: Trade-off between the tracking accuracy (pixel error) and efficiency (average computation time for processing each 1-second data stream) for saccade and smooth pursuit in EV-Eye.

4.4.2 Evaluation results on OpenEDS dataset

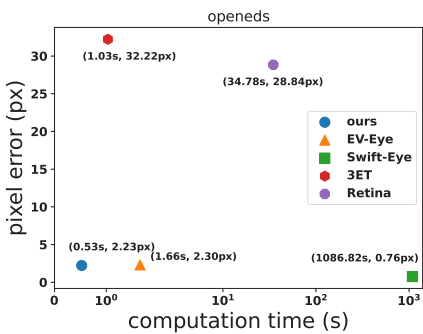


Fig. 10: Trade-off between the tracking accuracy (pixel error) and efficiency (average computation time for processing each 1-second data stream) in OpenEDS dataset.

To validate the efficacy of EX-Gaze in a VR environment, we compared it with four state-of-the-art event-based eye tracking methods using the event-based eye tracking dataset synthesized from OpenEDS, which was originally collected using VR headsets. To further demonstrate the generalization capability cross datasets, the models trained on the EV-Eye dataset were applied directly to the OpenEDS dataset

without any fine-tuning. Additionally, the thresholds TH_{ec} and TH_s were kept the same as those obtained in the ablation study using the EV-Eye dataset. Since the OpenEDS dataset does not contain labels specifying the types of eye movements, such as saccade or smooth pursuit, we are unable to provide detailed evaluation results based on these specific movement types, as was done with the EV-Eye dataset.

Figure 10 illustrates the trade-off between tracking accuracy and efficiency, showing the average pixel error against the average computation time for processing each 1-second data stream at a target tracking frequency of 2KHz. The results indicate that, even without further fine-tuning, EX-Gaze applied directly to the OpenEDS dataset achieves tracking accuracy comparable to more precise but computationally prohibitive methods. These alternative approaches fail to meet the real-time requirement of processing each 1-second data stream within 1 second at 2KHz. This demonstrates the effectiveness of EX-Gaze for realtime and high-frequency eye tracking with high-precision VR environments.

4.5 System Overhead Evaluation

To verify the real-time performance of our method, we evaluated the runtime of each module on a Jetson Orin Nano, with results summarized in Table 4. For a 2kHz tracking frequency, the computation for each cycle must be completed within 0.5ms to prevent accumulated latency. The event-based module achieves an average runtime of just 0.32ms, comfortably meeting the real-time requirement. While the frame-based relocalization module requires 1.17ms, its occasional use ensures that latency is mitigated by the high-speed event-based tracking process. As shown in Table 2 (Section 4.3.2), only approximately 8% of frames

Component	Frame-based relocalization	Event-based tracking	Event-frame Adaption	Gaze regression
Runtime	1.17ms	0.32ms	0.02ms	0.00012ms

Table 4: Module Overhead

undergo relocalization when a threshold of 0.8 is applied. Consequently, the estimated average runtime per tracking operation is about 0.407ms, satisfying the real-time requirement. Notably, these estimates exclude potential event-based tracking skips, which could further diminish the impact of frame-based latency.

5 CONCLUSION

In this paper, we introduce EX-Gaze, an event-based, real-time, and high-frequency eye tracking system designed for on-device extended reality applications, which are inherently resource-constrained. To leverage the sparsity of the event-stream, we propose a sparse event-patch transformer to achieve efficient eye tracking. Additionally, we design a lightweight event-frame approach adaptation scheme to guide the system's switch between frame-based and event-based approaches, thus avoiding costly frame-based relocalization while preserving high tracking accuracy. Through our evaluation and comparison with state-of-the-art methods, EX-Gaze demonstrates the best trade-off between tracking accuracy and efficiency, outperforming existing approaches by a significant margin.

Limitations and Future Work. It is important to note that our proposed method still has limitations. We have not yet built a prototype to validate the method in real-world scenarios, and even with synthetic datasets, we cannot fully replicate the complexities of real-world environments. In future work, we will construct a VR headset equipped with an event camera to evaluate the effectiveness of our method under various lighting conditions and user movement scenarios during real-world use. Moreover, our current method is focused solely on 2D gaze estimation. In the future, we will delve deeper into extending this method to 3D gaze estimation to further expand its application scope and improve performance.

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